ElasticBLAST: Using the power of the cloud to speed up science

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What is ElasticBLAST?

It is a command line tool to facilitate aligning large volumes of nucleotide and protein sequences against popular and user-provided BLAST databases. These cannot be done in NCBI WebBLAST or easily with BLAST+. ElasticBLAST is under active development.

When to use ElasticBLAST?

You can use ElasticBLAST anytime your search is likely to take more than an hour or two or your machine does not have enough resources to run BLAST.

An example application for ElasticBLAST is transcriptome annotation. Annotating transcripts with functional and biological processes is a critical step in developing an understanding of the biological complexity of an organism. This process typically involves querying multiple biological sequence databases and integration of complex software tools.

BLAST is typically used to align assembled transcriptome against well annotated databases of proteins or nucleotides to identify similarity and infer function. BLAST alignments help to find homologous genes, proteins, and functional domains that can be used as a basis for new annotation on the transcriptome.

An example annotation project is that of Opuntia streptacantha, commonly known as prickly pear or nopal (see NCBI BioProject ID PRJNA320545 and figure 1). Its transcriptome comprises 474,563 sequences and 217 million base pairs.

Figure 1: Opuntia streptacantha

Why should I use ElasticBLAST?

Here are a few benefits of using ElasticBLAST:

- Access virtually unlimited compute and storage capacity
- Pay-as-you-go model
  - you only pay for what you use
  - No upfront large capital expenditures
- Reduce the learning curve to be effective in the cloud
  - Easily run BLAST on the cloud
  - No need to own and maintain hardware
  - No need to store and update copies of NCBI BLAST databases
  - ElasticBLAST aims to hide the complexities inherent to working in the cloud

ElasticBLAST features

Below are a few highlights of ElasticBLAST's features:

- Supports all BLAST programs: blast[pnx], tblast[nx], megablast, psiblast, rpsblast, rpstblastn.
- Supports user-provided and a growing list of NCBI BLAST databases.
- Supports taxonomic filtering of BLAST databases (both inclusion and exclusion of taxonomic groups).
- The ElasticBLAST source code and its documentation are freely available.
- Runs on Amazon Web Services (AWS) and the Google Cloud Platform (GCP).
- Supports horizontal auto-scaling: the ability to increase the number of compute instances automatically, up to a pre-defined amount, based on its workload (see figure 2).
- Supports spot/preemptible instances, which allows ElasticBLAST users to tap into the Cloud Service Provider's excess compute capacity at a discounted price.

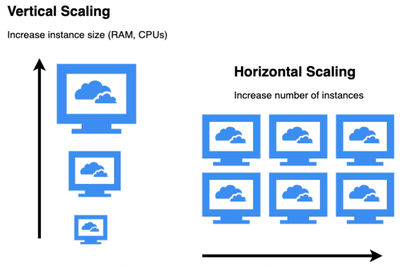

Figure 2: Types of scaling in the cloud

Underlying technologies

- Docker: for using containers with BLAST+, GCP and AWS Software Development Kits (SDKs). Using containers facilitates dependency management and reproducibility.
- Cloud object storage: Simple Storage Service (S3) in AWS and Google Cloud Storage (GCS) in GCP. Used to store BLAST databases, queries, results and ancillary data used during the ElasticBLAST execution.
- Kubernetes in GCP and AWS Batch in AWS for cloud resource management and job orchestration.

Architecture and workflow

The following descriptions focuses on the cloud operations of ElasticBLAST. The client refers to the machine that runs ElasticBLAST (it could be a machine inside or outside the cloud).

Please refer to figure 4 for the steps in this paragraph, which focuses on AWS. The first step involves creating cloud resources, which is accomplished via AWS CloudFormation, an infrastructure-as-code service that creates "stacks" of resources built from a "template". Step 2 involves job submission to AWS Batch, which schedules the jobs (step 3). Once the jobs start to execute (step 4) they download the query batches, BLAST database and run BLAST. Once BLAST completes its execution, the results are written onto the user's results bucket (step 5). To finalize processing, the client invokes the ElasticBLAST delete operation to release all cloud resources and continue to process the BLAST results (step 6).

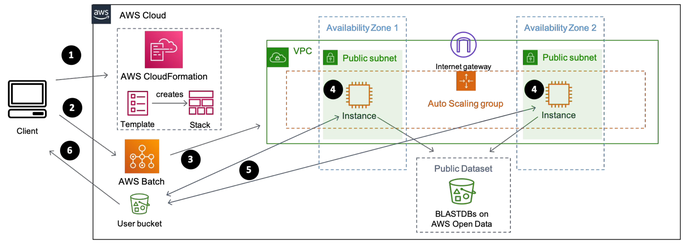

Figure 4: Architecture and workflow overview on AWS

In ElasticBLAST on GCP (see figure 5), the client creates a Kubernetes cluster (step 1), submits a job to initialize a shared disk with query batches and BLAST database (step 2), and Kubernetes schedules and runs it (step 3). The client then submits jobs to run BLAST (step 4) and Kubernetes again handles these jobs (step 5). After BLAST completes, its results are written to the user's results bucket (step 6) and the user invokes ElasticBLAST delete to release all cloud resources and continue to process the BLAST results (step 7).

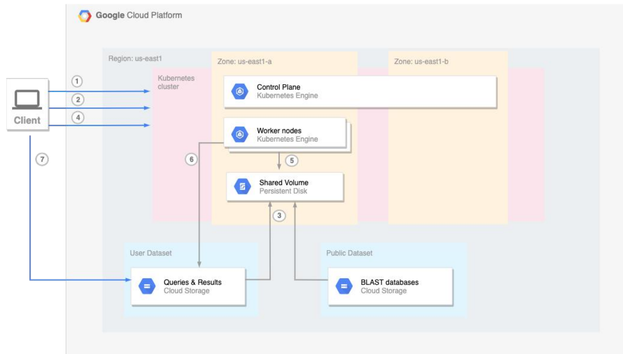

Figure 5: Architecture and workflow overview on GCP

Cost and runtime performance

Figure 6 compares the cost and runtime for BLAST+ running on a single server and ElasticBLAST running on 10 servers to align the aforementioned Opuntia streptacantha transcriptome against the nt BLAST database using megablast with default parameters. Using ElasticBLAST one can do this BLAST search 7 to 11 times faster than it is possible on a single server using BLAST+. Using spot instances may result in longer runtimes than using on-demand instances, but as illustrated in the example below it can yield significant cost savings.

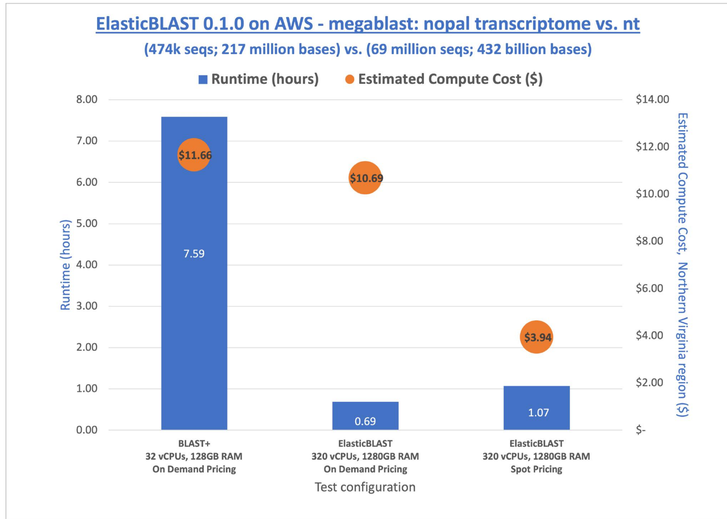

Figure 6: Sample cost and runtime performance

Availability

- Source code: https://github.com/ncbi/elastic-blast
- Documentation: https://blast.ncbi.nlm.nih.gov/doc/elastic-blast
- PyPI: https://pypi.org/project/elastic-blast/
- BioConda: https://anaconda.org/bioconda/elastic-blast

Future work

- Automatic shutdown of all cloud resources upon completion.
- Job submission on the cloud.
- Performance optimizations.
- Publication.

Want to try ElasticBLAST?

NCBI is looking for alpha testers to try ElasticBLAST, provide feedback and help guide its future developments.

If you have feedback or want to chat with us, please email us at elastic-blast-support@ncbi.nlm.nih.gov

Join us at the BOSC Collaboration Fest on July 31-August 1, 2021!

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